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DIGITAL ARCHITECTURE AND REFORMING THE BUILT ENVIRONMENT

Rabee M. Reffat

Digital technologies are undoubtedly affecting architecture and the built environment in unprecedented ways. In digitally designed architecture, computational design models capable of dynamic transformation are replacing the static norms of conventional processes favoring computationally generated complexities. This paper focuses on some of the reformations in the built environment from the epistemological lens of digital architecture, including the role that digital architecture might have on the types and forms of buildings and architectural spaces in the built environment. The manifestations of digital architecture on reforming the built environment, including the new paradigm of architectural forms, narrative and interactive spaces, digital tectonics, and digital fabrication, are articulated in this paper.

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INTRODUCTION

Computing has become one of the most important transformations of contemporary professions, including architectural education and practice. Information technologies and the construction industry are coming together in ways once unimaginable, and their union is changing what people do inside buildings, as well as how buildings are designed and built. For the last two centuries, the basic construction paradigm has been the factory, created for masses of people to work together to make complicated products. That paradigm has been applied not only to the workplace but also to homes, schools, and most other types of buildings. The industrial society has created high-rise apartment buildings as human warehouses so that people can live near their workplace. Even the interior design of homes has been influenced, subdivided like factories into a series of special purpose rooms, e.g., kitchen, dining room, living room, television room, sewing room, and bedroom (see also Zwarts and Coolen, 2006). Now that paradigm is changing with the rise of the virtual workplace, and physical proximity is becoming ever less essential. The built environment of office buildings where people work, houses where they live, factories that manufacture their goods, and malls where they shop are on the threshold of dramatic change driven by advances in digital computer and communication technologies. The new move is supposed to free people from the constraints of place. Geography no longer matters. With everyone wired digitally, one could work, socialize, and entertain at and from anywhere. Therefore, it is important to examine the computer's cognitive implications for architecture and whether there are emerging new modes of thinking about architecture, space, and form of the built environment.

The most direct manner through which architecture presents itself is by means of spatial form. When the process of architectural production undergoes change, then changes in the form of architecture follow. A combination of structure and materials moves the power of architecture to become a real and existing building (Giedion, 1967). The development of information technology and its applications to design disciplines have changed how buildings are described and even how they are built. Engineers, architects, contractors, and other parties often exchange files instead of paper drawings, and manufacturers can be sent numeric data to guide the fabrication of customized building components. The computer has gone from being an isolated box to become part of a gigantic digital network of networks that shapes our collective future. The way and pace at which we connect, communicate, memorize, imagine, and control the flows of valuable information have changed dramatically. The digital phenomena that directly affect the architectural world include miniaturization, ubiquity, real-time, networked, and virtuality. Temporal contiguity and temporal connectivity have taken precedence over spatial and geographical contiguity. These phenomena have squeezed, stretched, restructured, reconfigured, and redistributed most major human institutions. Consequently, the built environment's role, importance, and nature have changed. The new measures of architecture are connectivity and speed. The architects of a new world need to recognize these transformations and think differently (Senagala, 2003).

Architecture has undergone an important transition. After a long period of contestation, the digital has emerged the undisputed victor in the struggle for the core of architectonic conception and execution. Architecture is enjoying a renaissance, and the most vibrant work being done today would be unthinkable without computers. Computers have created a direct link between what can be conceived and what can be built, augmenting both. Architects are finally learning to design by algorithm, however, it is not always so (Novak, n.d.).

DIGITAL AFFORDANCES

Buildings in the built environment change in response to every major addition to the palette of options available to architects. This has been witnessed after the development of new materials (iron, steel, concrete, and glass) and new technologies (electrical lighting, elevators, and air conditioning). The full potential of the technological change is not felt until designers begin to explore the ways in which the new technology allows them to think differently about buildings. At that point, architectural expression often undergoes a significant change at levels ranging from tectonics to building types. Information technologies (IT) are no different. Exploration of the architecture enabled by information technology allows ex-

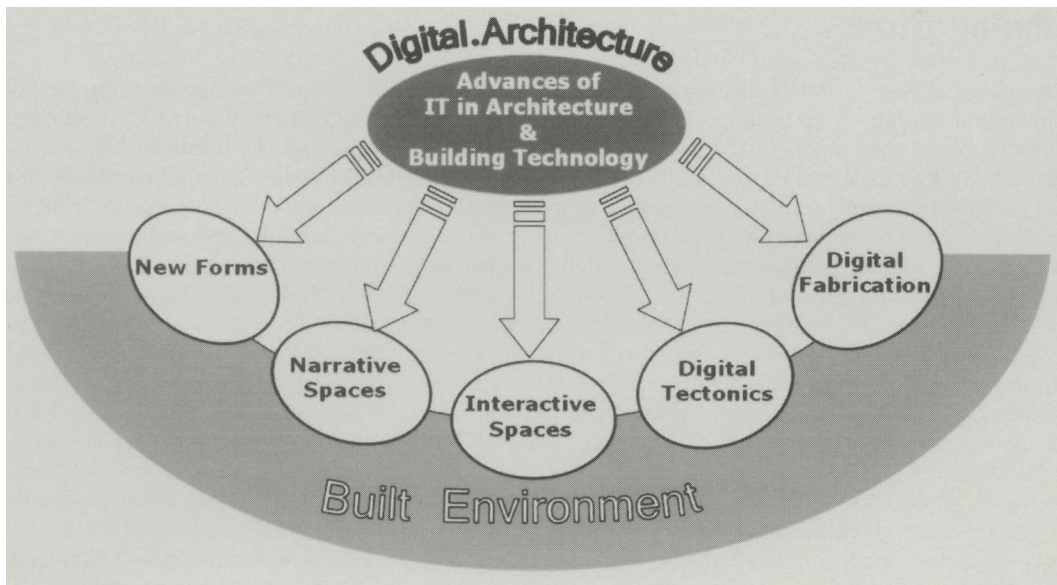


FIGURE 1. Some of the major impacts of digital architecture on reforming the built environment.

tending the design paradigm of traditional architecture into the manufacturing process. Design expands from a process of setting geometry, assembling pre-existing building systems, and specifying finishes, growing to take on the specification and design of the systems themselves, often working with the expertise of the manufacturer. This approach offers the designer greater morphological freedom and it may reverse the erosion of the scope of professional services by expanding the province of the architect. The recasting of the process into a digital framework creates digital affordances or opportunities that are not found in the traditional circumstances (Johnson, 2000).

The Information Age, like the Industrial Age before it, is challenging not only how buildings are designed but also how they are manufactured and constructed. The guiding notion is that generative and creative potential of digital media, together with manufacturing advances already attained in automotive, aerospace, and shipbuilding industries, is opening up new dimensions in architectural design by allowing production and construction of very complex forms that were until recently very difficult and expensive to design, produce, and assemble using traditional construction technologies. The proposition is that the consequences of these changes are likely to be profound, as new digitally driven processes of design, fabrication, and construction are increasingly challenging the historic relationship between architecture and its means of production (Kolarevic, 2001).

DIGITAL ARCHITECTURE AND REFORMING THE BUILT ENVIRONMENT

Tremendous developments have occurred during the last decades of the 20th century through the digital age, including information technology and communication. The digital age has influential impacts on most living activities. The impacts of fast and progressive developments of information technology manifested in diverse computational processes of architectural designing resulted in digital architecture, along with the advances of building technologies and manufacturing that has been increasingly witnessed in the past two decades. The dimensions of these impacts on the built environment are too vast to be encompassed within the scope of this paper. This paper focuses on some of the major impacts of digital architecture on reforming the built environment, as depicted in Figure 1, including (1) a new paradigm of architectural forms, (2) narrative and interactive spaces, (3) digital tectonics, and (4) digital fabrication. These impacts are articulated in the following subsections.

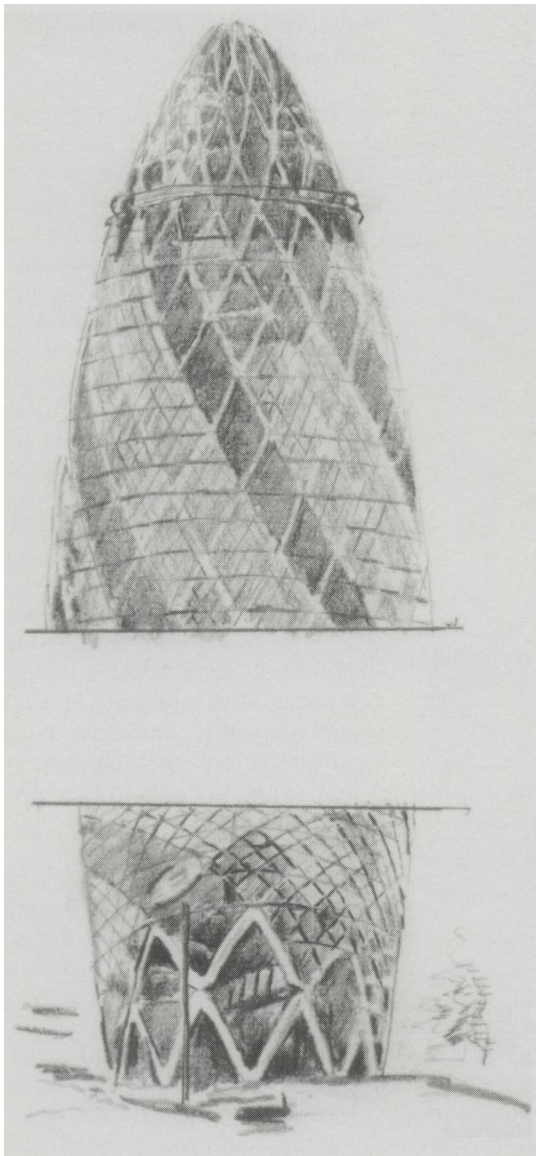


FIGURE 2A. The digital architecture reshaping building forms in the built environment: Swiss Re Building, London.

A New Paradigm of Architectural Forms

Digital architecture with its new representational technology affects the way designers work in terms of speed, as well as promoting the use of static elements (paper, plastics) and of such notions as time, motion, and flow, through to the generation of simulations that can be used to work differently, to imagine space, and to relate to it in a new way (see also Park, 2006). This is dramatically changing the conventional approach to design. It also allows architects to work with a vector geometry which marks the end of the rule of the Euclidean geometry. This breaking off involves designers' conceptual and designing potentials as much as their implementation. Increased production, reduced designing costs, improved communication of the executive building choices, and a new conception of the architectural space are just some of the major consequences of the use of new technology in architectural design. In the digital architecture paradigm, coatings are turned into screens on which pictures flicker (a blend of architecture and media). Also, the dynamics of architectural space and the building envelope as its expression appear to have reached a current state of flexibility due to the achievements of modern instruments of computation and virtual simulation. This is a paradigm of new digital message that comes to assume theoretical bases of fractal computation towards the complexity of the nonlinear dynamics generated by the meeting of forces and resistances with the flow of climatic factors coming from the outside. For instance, in the Swiss Re Building, London, as shown in Figure 2A, the envelope is broken down and recomposed in a three-dimensional space; framed by a fresh variety of viewpoints; dematerialized by plastic and curvilinear shapes, by the subtraction of parts, by an almost evanescent transparency that invites experimentation in crossing it; and brought to the most contemporary fluid deformations of the archi-

tectural material until the expression of an almost vital dynamism. It is an extensive diagram in which vector signs and systematic lines of force can be reduced to synthesis, which allows controlling the manipulation of the shapes under light, the increases in scale, the combination of viewpoints, the movement, and the stagnation (Fiamma, 2003). Dynamic architecture means flexible architecture that can change over time. The dynamics also applied to the creative process behind the architectonic object, employing 3D animation software to create forms that incorporate a sense of time, the concepts of bubbles and waves have digitally formed both BMW IAA Pavilion, Frankfurt, and BMW Clean Energy Expo, Hanover, as illustrated in Figure 2B. The architectural highlight in the exposition hall was a vaulting "wave" made of an aluminum tube spanned with translucent projection foil. The background showed a "sun" made of a steel frame construction and translucent spanned foil. Beneath the "wave" spanned light walls split the space which observers could walk in.

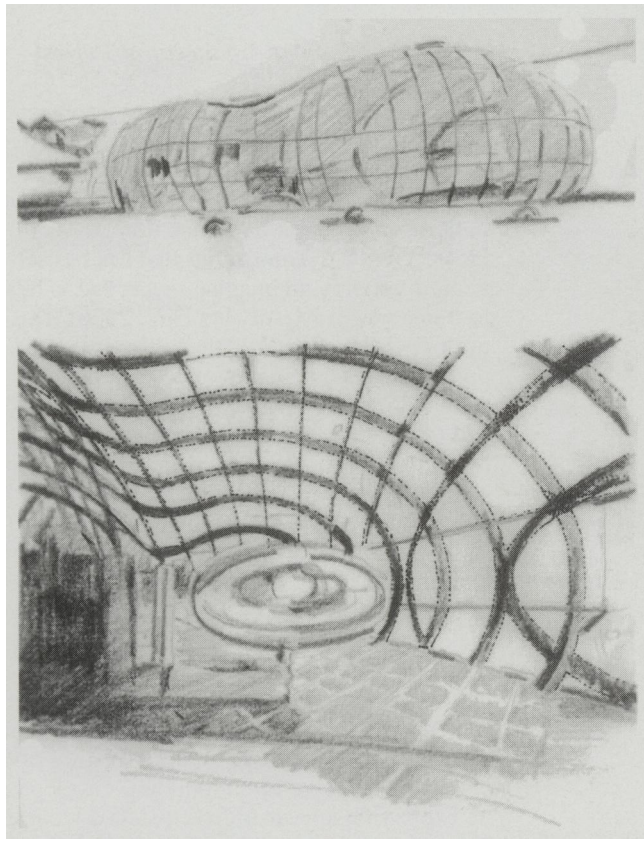


FIGURE 2B. The digital architecture reshaping building forms in the built environment.
Top: BMW IAA Pavilion, Frankfurt; bottom: BMW Clean Energy Expo, Hanover, Germany.

Narrative and Interactive Spaces Reshaping Cities

Architecture is no longer simply the play of masses in light. It now embraces the play of digital information in space. We will characterize cities of the 21st century as systems of interlinked, interacting, silicon- and software-saturated smart, attentive, and responsive places. We will encounter them at the scale of clothing, rooms, buildings, campuses, neighborhoods, metropolitan regions, and global infrastructures (Mitchell, 1999). Computers, the new digital media, liberate the duality of concepts of space in human civilization. The construction and simulation powers of digital media trigger all kinds of unlimited imagination. The new space of this kind may be called digital space or virtual space. This new space is between mental and physical spaces because it provides designers with not only unlimited imagination of mental space but also live-inside perception of physical space. A new concept of space of mankind is thus generated (Liu, 2003).

People experience their lives as a narrative. Bruner (1990) stressed the importance of story as the means that structures our perception and communication. Narrative is our main instrument

of making meaning, the embodiment of culture, communication, and education. The history of architecture offers innumerable examples of places that embed and narrate a story through their spatial layout and décor. By looking at the sequence of floor plans of historical buildings through the centuries, from the Greek temple, to the Roman church, the medieval dome through today, one can understand how a rectangle, a circle, a cross, or other more complex figures transmit a message through the centuries. This message is a story about how people through times relate to life, nature, and spirituality. The new information society architecture can more explicitly embed stories and information in its structure due to the tools offered to us by the digital revolution. In the last decade, the architectural landscape of the contemporary city has been undergoing a profound transformation. In today's cities, large billboards with large LCD screens show us publicity, entice us, talk to us, and inform us about our banking, clothing, insurance, entertainment, and eating options. These billboards are embedded in the city architecture and have become integrated in the visual appearance and profile of the cityscape (Sparacino, 2002). Times Square in New York, as shown in Figure 3, is one example of many narrative spaces evidently visible in today's cities.

The interactivity between buildings, city spaces, and their users will depend on the purpose of spaces and their different roles in the landscape or townscape. Architectural spaces might have large projection surfaces with commercials, information, and entertainment. They might support communication to mobile devices in the neighborhood. Such communication may be information about activities in these buildings or open spaces, how to find your way in the building, or information about other services provided by the



FIGURE 3. Narrative spaces in Times Square, New York.

building or the organizations it hosts. Such services to mobile devices are a special case of location-based services, where users of mobile devices may receive information based on their physical location.

Digital models of buildings and cities can also be used to make the living and democratic processes in cities more interactive and tangible. An example of this is the Karlskrona2 project. The project is to create a digital copy of the city Karlskrona in Sweden, to be called Karlskrona2, which will be accessible to the citizens of the city via the Internet. Digital representations of themselves (avatars), the streets, and buildings of the city center will create a virtual version of the structure and personal relationships in Karlskrona (the real city). Karlskrona2 will initially be an exact replica of the city, but as the virtual citizens meet and interact, things will change, buildings will redefine their function, social hierarchies will alter, and laws will be reconstituted and renewed. The virtual Karlskrona will be visible to the real city through a large-scale video projection in the main square. Citizens can gather in real space to follow the activities of their avatars and consider the divergences between Karlskrona and Karlskrona2 (Karlskrona2, n.d.). The interaction between virtuality and reality of Karlskrona and Karlskrona2 is shown in Figure 4.

The digital mirror of Karlskrona is accessible from web browsers, large projection screens, and kiosks in the city. Citizens may discuss changes to the city through concrete proposals in terms of model elements entered into the digital mirror of Karlskrona. Such proposals can then be discussed via a kind of 3D chat environment. In this way, citizens in the real city may also become citizens in a proposed virtual city, where an ongoing debate about the real city takes place. Some types of buildings, e.g., supermarkets and manufacturing plants, play special roles in relation to transportation and logistics in the environment. Drivers who need to deliver goods need to be able to get information about access conditions, preferred delivery entrance, other trucks in line, etc. Building information systems and transport information systems need to be integrated, and the domain of intelligent transport has become a focus area in today's cities (Krogh and Grønbæk, 2001).



FIGURE 4. Interaction between virtuality and reality: projections of the Karlskrona2 project (virtuality) in the mail square, Karlskrona in Sweden (reality).

Digital Tectonics

In the conceptual realm, digital architectures of topological, non-Euclidean geometric space, kinetic and dynamic systems, and genetic algorithms are supplanting technological architectures. Digitally driven design processes characterized by dynamic, open-ended, and unpredictable, but consistent, transformations of three-dimensional structures are giving rise to new architectonic possibilities (Kolarevic, 2001). Architectural tectonics commonly refers to focusing on structural clarity, materiality, and attention to detail while assembling building components in architecture. The industrial age brought with it the realization that buildings can be thought of and assembled in similar ways to the industrial artifacts and machinery that were being manufactured. One of the earliest and largest examples of a building that was thought of and assembled as a kit-of-parts is Sir Joseph Paxton's 1851 Crystal Palace project in London (Jabi, 2004). Digital tectonics has been converged as a term to symbolize the poetics of digitally conceived, structurally and materially clarified, and directly manufactured architecture (Beesley and Seebohm, 2000). Leach, *et al.* (2004) describe the term digital tectonics as a new paradigm of thinking in architectural culture facilitated by, but not totally dependent upon, the technological possibilities afforded by the digital realm. The shift to digital tectonics has been described as a "structural turn" due to a renewed focus on the structural integrity of buildings and new dialogues and collaborations between architects and engineers.

The use of digital tools has made a noticeable impact on architecture as a whole and architectural tectonics in particular. For instance, in the work of Frank Gehry, the utilization of digital tools allowed a more accurate translation of architectural ideas into built artifacts. It is an apparent transformation of architecture as a set of assembled components into a set of more homogenous and reductionist volumes and surfaces. This transformation can be exemplified if one compares Frank Gehry's early projects that make use of digital tools, such as the Fred and Ginger building in Prague, Czech Republic (shown in Figure 5A), to more recent projects such as the Disney Concert Hall in Los Angeles (shown in Figure 5B). The Fred and Ginger building, while bearing the mark of digital design in its deformed surfaces and non-repetitive paneling geometry, continues to evoke the tectonic in its materiality and assemblages. In contrast, Disney



FIGURE 5A. Fred and Ginger building in Prague, Czech Republic (Dancing Building), 1995, Frank O. Gehry and Vladimír Miluněk.

Concert Hall conceals its structural systems and building components to present to the viewer an abstract sculpture, as if made of a single sheet of metal that has been made to deform according to the will of the architect (Jabi, 2004).

Digital Fabrication

Integrating computer-aided design with computer-aided fabrication and construction fundamentally redefines the relationship between designing and producing. It eliminates many geometric constraints imposed by traditional drawing and production processes, making complex curved shapes much easier to handle, for example, and reducing dependence on standard, mass-produced components. It bridges the gap between designing and producing that opened up when designers began to make drawings. By integrating design, analysis, manufacturing, and assembly of buildings around digital technologies, architects, engineers, and builders have the opportunity to reinvent the role of a master builder and reintegrate the currently separate disciplines of architecture, engineering, and construction into a relatively seamless digital collaborative enterprise, thus bridging the gap between designing and producing that opened up when designers began to make drawings (Mitchell and McCullough, 1995).

Architecture continually informs and is informed by its modes of representation, perhaps never more so than now, when digital media is rapidly expanding what we conceive to be formally, spatially, and materially possible. The increase in the use of the computer has led to the claim that the body is becoming increasingly distant from architectural production. However, as architects engage computerized input and output technologies, opportunities arise for greater confluence and specificity between the embodied material world, immaterial phenomena, and architectural space. Pedagogically, the immediate tangibility of the body offers valuable and inescapable geometries that both push and draw upon digital methods of generating supple form. Experientially, it affords investigation and critique of something intimate and familiar (Iwamoto, 2004). The potential of digital media associated with manufacturing advances is opening up new dimensions for digital fabrication of architectural designs. Zellner (1999) observed that architecture is recasting itself, becoming in part an experimental investigation of topological geometries, partly a computational orchestration of robotic material production and partly a generative and kinematic sculpting of space.

Architects are having an increasing interest in new materials with the diffusion of digital processes of conception and production. Such interest stems from the new geometric complexities especially in surface or shell structures in which the skin absorbs all or most of the stresses. This will blend the structure and the skin into one element to create self-supporting forms. Therefore, a search has been promoted for new materials, such as high-temperature foams, rubbers, plastics, and composites, which until recently were

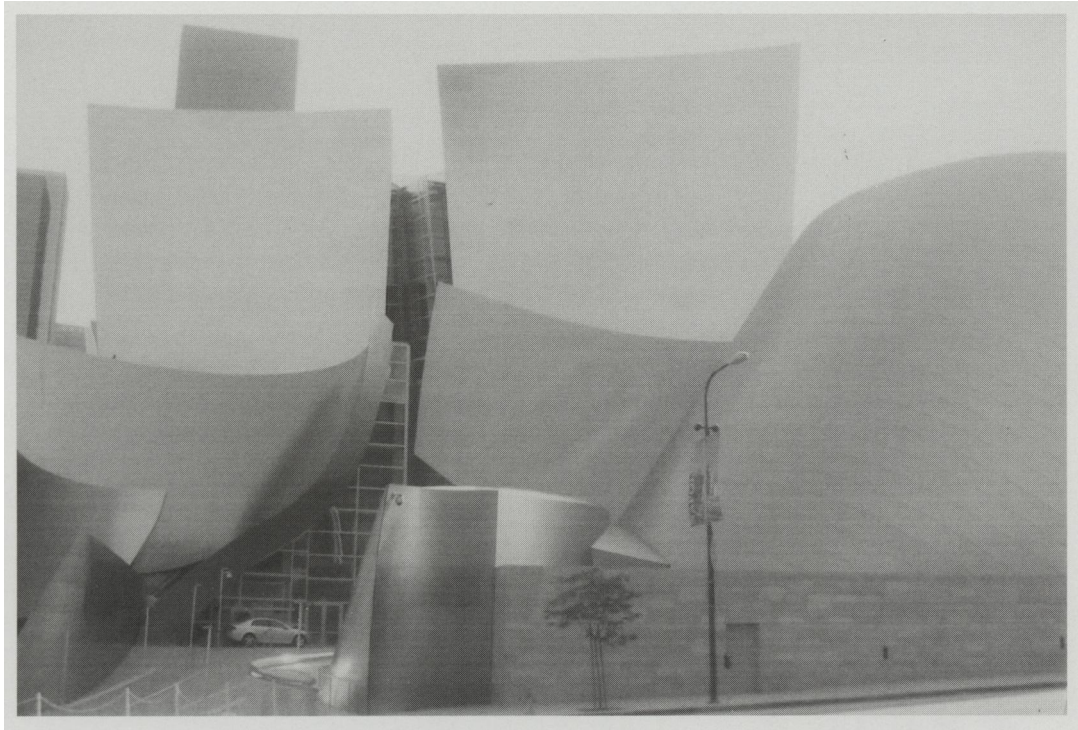


FIGURE 5B. Exterior view, Disney Concert Hall, Los Angeles, 2004, Frank O. Gehry.

rarely used in the building industry. The idea of a structural skin not only implies a new material but also geometries, such as curves and folds that would enable the continuous skin to act structurally, obviating an independent static system. The skin alone does the heavy lifting. The implications of the new materiality are significant and promise a radical departure from Modernism's ideals. In some ways the search for a material and form that unifies structure and skin is a counterrevolution to Le Corbusier's Domino House, in which the master separated structure from skin. The new conflation is a return to the bearing wall, but one with freedom that Le Corbusier never imagined possible. Complex surfaces with integrated structures promise a quantum leap of engineering elegance and intellectual satisfaction (Giovannini, 2000; Kolarevic, 2001).

Mass customization can be defined as mass production of individually customized goods and services (Pine, 1993). Mass customization offers a tremendous increase in variety and customization without a corresponding increase in costs. The ability to mass-produce irregular building components with the same facility as standardized parts introduced the notion of mass customization into building design and production. Nowadays, it is just as easy and cost-effective for a CNC (Computer Numerical Control) milling machine to produce thousands of unique objects as to produce thousands of identical ones. CNC mills can perform the functions of drilling and often turning. CNC-driven production processes afford the fabrication of non-standardized repetitive components directly from digital data. That is precisely what Greg Lynn's "Embryologic Houses" manifest, "mass-customized" individual house designs produced by differentiation achieved through parametric variation in non-linear dynamic processes (Kolarevic, 2001). Embryologic houses are a line of one-of-a-kind housing and a strategy for the invention of domestic space. The embryologic houses employ a rigorous system of geometrical limits that liberate an exfoliation of endless variations. This provides a generic sensibility common to any embryologic house at the same time that no two houses are ever identical.

CONCLUSION

Digital technologies are undoubtedly affecting architecture and the built environment in unprecedented ways. Today we are at the early stages of what will be seen as a critical moment in the history of these technologies and of their impact on all aspects of spatial theory. The use of digital technologies has spawned various strategies and deployments. It is evident today that some architects are searching for new forms, some for new techniques, while still others are seeking out new territories. Computers and the enhanced capabilities they afford have become essential admirers in the future course of architectural research and discourse.

The new materials, methods, and theories engendered by digital technologies will continue influencing and changing the ways in which architects think and work. Well-established architectural conventions are currently displaced by the digital paradigm shifts witnessed in contemporary and digital architectural designs. In digitally designed architecture, computational design models capable of dynamic transformation are replacing the static norms of conventional processes favoring computationally generated complexities. With digital architecture, the relationship between architecture and its means of production is being challenged by the digital processes of design and fabrication.

The buildings we design and the designs that we build shape and inflect who we are, how we think, and the kind of reality we construct as individuals and as a culture. The conventional processes of designing the built environment are simply challenged by the current generation of digitally oriented architects. For better or worse, this is the shape and texture of digital architecture in the digital age. In addition, it should come as no surprise that it reforms the built environment in its form, shape, and activities. Since information technology is complex and rapidly changing, the built environment will continually be reformed, and its architecture will change.

REFERENCES

- Beesley P, Seebohm T (2000) Digital tectonic design. In D Donath (Ed.), *Promise and reality: State of the art versus state of practice in computing for the design and planning process* (Proceedings of the 18th eCAADe Conference). Weimar, Germany: Bauhaus-Universitat, pp. 287–290.
- Bruner J (1990) *Acts of meaning*. Cambridge: Harvard University Press.
- Fiamma P (2003) Architectural design and digital paradigm: From Renaissance models to digital architecture. In K Klinger (Ed.), *Connecting crossroads of digital discourse* (Proceedings of the 22nd Annual Conference of the Association for Computer-Aided Design in Architecture). Indianapolis: ACADIA, pp. 247–253.
- Giedion S (1967) *Space, time and architecture — The growth of a new tradition*. Cambridge: Harvard University Press.
- Giovannini J (2000) Building a better blob. *Architecture* 89(9):126.
- Iwamoto L (2004) Embodied fabrication: Computer aided space-making. In P Beesley, NY Cheng, and RS Williamson (Eds.), *Fabrication: Examining the digital practice of architecture* (Proceedings of the 23rd Annual Conference of the Association for Computer-Aided Design in Architecture). Ontario: ACADIA, pp. 270–281.
- Jabi W (2004) Digital tectonics: The intersection of the physical and the virtual. In P Beesley, NY Cheng, and RS Williamson (Eds.), *Fabrication: Examining the digital practice of architecture* (Proceedings of the 23rd Annual Conference of the Association for Computer-Aided Design in Architecture). Ontario: ACADIA, pp. 256–269.

- Johnson S (2000) Binary oppositions: Should designers learn to think differently in order to better utilize digital design tools? *ACADIA Quarterly* 19(4):2.
- Karlskrona2 (n.d.) <http://www.superflex.net/tools/supercity/karlskrona2/index.shtml>. Site accessed March 2006.
- Kolarevic B (2001) Digital fabrication: Manufacturing architecture in the information age. In W Jabi (Ed.), *Reinventing the discourse — how digital tools help bridge and transform research, education and practice in architecture* (Proceedings of the 21st Annual Conference of the Association for Computer-Aided Design in Architecture). New York: ACADIA, pp. 268-277.
- Krogh P, Grønbæk K (2001) Architecture and pervasive computing — When buildings and design artifacts become computer interfaces. *Nordic Journal of Architectural Research* 14(3):11.
- Leach N, Turnbull D, Williams C (2004) *Digital tectonics*. Hoboken, NJ: Wiley-Academy Press.
- Liu YT (2003) Digital architecture: Theory, media and design. In ML Chiu, JY Tsou, T Kvan, M Morozumi, and TS Jeng (Eds.), *Digital design research and practice* (Proceedings of the 10th International Conference on Computer-Aided Architectural Design Futures). Tainan: Kluwer Academic Publishers, pp. 9-18.
- Mitchell WJ (1999) *E-topia: Urban life, Jim — but not as we know it*. Cambridge: MIT Press.
- Mitchell WJ, McCullough M (1995) *Digital design media*, 2nd ed. New York: Van Nostrand Reinhold.
- Novak M (n.d.) Liquid, trans, invisible: The ascent and speciation of the digital in architecture, a story. <http://www.a-matter.de/digital-real/eng/main.asp?es=6>. Site accessed March 2006.
- Park JH (2006) Arraying alternative archetypes: A computational model with a Java Applet. *Journal of Architectural and Planning Research* 23(3):235-246.
- Pine BJ (1993) *Mass customization: The new frontier in business competition*. Boston: Harvard Business School Press.
- Senagala M (2003) Digital theory. In K Klinger (Ed.), *Connecting crossroads of digital discourse* (Proceedings of the 22nd Annual Conference of the Association for Computer-Aided Design in Architecture). Indianapolis: ACADIA, pp. 254-255.
- Sparacino F (2002) Narrative spaces: Bridging architecture and entertainment via interactive technology. In C Soddu (Ed.), *The 6th International Conference on Generative Art*. Milan, Italy: Generative Design Lab, DiAP, Politecnico di Milano University.
- Zellner P (1999) *Hybrid space: New forms in digital architecture*. New York: Rizzoli.
- Zwarts A, Coolen H (2006) The meaning of preferences for residential environment features: A case study among apartment dwellers in the Netherlands. *Journal of Architectural and Planning Research* 23(3):200-215.

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